

M03 Protected Review & Repair

OPEN ONLY AFTER the matching genuine attempt is saved. A corrected answer after reading review is learning evidence, not mastery. Close review and perform a changed retry if the first attempt was materially weak.

R-DE03-01 - INNER JOIN and LEFT JOIN

Choose join type from business eligibility. Check unmatched rows explicitly. A 1-side key should be proven unique before assuming one order remains one row.

Reviewer check: No non-West/non-Completed rows in the first query. Second query preserves the Completed order count; returned orders have return_id and others are NULL.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-02 - Cardinality, row multiplication and join validation

The raw order-to-contact LEFT JOIN produces 35 rows from 30 orders because several customers have multiple contact rows. Preserve order grain by summarizing/selecting contacts first when order-level output is required.

Reviewer check: Output key order_id remains unique; row count is 30; contact_count can exceed 1 without duplicating orders.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-03 - UNION and UNION ALL

UNION ALL preserves both streams. UNION additionally removes duplicate result rows. Align number/type/order of columns in both branches.

Reviewer check: You can explain exactly which duplicate combination UNION removed and why.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-04 - String functions for cleaning and standardization

Keep raw and transformed text comparable. TRIM/case formatting is a derived transformation; ambiguous business corrections need a rule/source.

Reviewer check: Raw and transformed names remain visible; the filter returns only names matching the intended text rule.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-05 - Dates, timestamps and time analysis

Validate date/timestamp types and timezone meaning. A calendar date has no timezone; event timestamps need a contract before cross-zone comparison.

Reviewer check: Daily groups reconcile back to the Completed net-sales total; timezone limitation is stated rather than guessed.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-06 - Subqueries and nested logic

Run the inner query alone first. Confirm scalar vs set semantics. Keep eligibility rules consistent when comparing an order/customer with an average or list.

Reviewer check: State inner and outer grain separately and reconcile customer totals to overall Completed net_sales.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-07 - CTEs and readable multi-step SQL

Name each CTE by purpose and state its grain. CTEs improve structure; they are not automatically faster.

Reviewer check: Every CTE has a written grain comment and totals reconcile.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-08 - Window functions and PARTITION BY

Window functions can keep detail rows. GROUP BY collapses. Validate window totals against a separate GROUP BY control.

Reviewer check: Detail row count stays 27; a separate GROUP BY customer query matches the window totals.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-09 - ROW_NUMBER, ranking and LAG/LEAD

Latest/top-N logic needs deterministic ORDER BY. Use PARTITION BY to reset sequence by entity/group; add stable tie-breakers.

Reviewer check: At most 2 rows/region; order ranking is reproducible; compare each region maximum manually/separately.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-10 - Advanced aggregation and multi-grain reasoning

Pre-aggregate each source at a compatible grain before joining. DISTINCT cannot repair multiplied SUMs. Reconcile each source metric independently.

Reviewer check: 4 final rows; each component total reconciles independently; document any customer-to-region assumption.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-11 - PostgreSQL setup, schemas and dialect awareness

PostgreSQL is a deliberate second engine. Confirm actual version/connection/database/schema. Do not confuse MySQL USE database syntax with PostgreSQL schema/search_path behavior.

Reviewer check: Query runs in PostgreSQL, not MySQL; result has 12 rows; dialect notes are evidence-based rather than invented.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-12 - Constraints and transactions

A constraint failure is evidence the rule worked. PostgreSQL may mark the current transaction aborted after an error; ROLLBACK before continuing. Never remove integrity just to silence an exercise.

Reviewer check: Baseline row count restored after rollback; no invalid row persists.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-13 - Indexes and EXPLAIN basics

On tiny tables, Seq Scan after creating an index can be rational. Report the actual plan; prove the index exists separately. EXPLAIN is plan evidence, not measured runtime proof.

Reviewer check: Evidence includes the actual index definition and plans; explanation does not equate "index exists" with "query is faster."

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-14 - Views and reusable database logic

A normal view stores reusable query logic and reads underlying data. It is not a materialized cache and does not automatically improve performance. Validate the view grain.

Reviewer check: One row/customer with activity; sum of gross_sales matches an independent aggregate over eligible source rows.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-15 - Integrated unseen SQL challenge

Strong work separates source grains, prevents fan-out, reconciles money/counts, and preserves independent PostgreSQL evidence. Repair only failed competencies; use fresh retest afterward.

Reviewer check: All monetary totals reconcile, customer_id is unique in final customer table, latest-ticket logic is deterministic, and PostgreSQL evidence is observed/saved.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.

R-DE03-16 - Timed SQL interview set and explanation

Score correctness + grain + validation + explanation. Timed mastery must use a new set after review, not memorized repetition.

Reviewer check: Save SQL, result controls, time used, confidence, errors and explanation. Review remains closed until the attempt is preserved.

Fresh-retry rule: change dataset/filter/entity/domain detail so the learner cannot reproduce the reviewed wording from memory.